

```

1 import serial
2 import time
3 import matplotlib.pyplot as plt
4
5 PORT = "COM3"
6 BAUD = 115200
7
8 ser = serial.Serial(PORT, BAUD, timeout=1)
9
10 times = []
11 values = []
12 start = time.time()
13
14 plt.ion()
15
16 fig, ax = plt.subplots()
17 line, = ax.plot([], [], marker="o", label="Napětí")
18
19 ax.set_title("Napětí v čase")
20 ax.set_xlabel("Čas [s]")
21 ax.set_ylabel("Napětí [V]")
22 ax.set_ylim(0, 20)
23 ax.grid(True)
24 ax.legend()
25
26 info_text = ax.text(
27     0.02,
28     0.95,
29     "",
30     transform=ax.transAxes,
31     verticalalignment="top"
32 )
33
34 print("Čtu data...")
35
36 while True:
37     raw = ser.readline().decode(errors="ignore").strip()
38
39     if not raw:
40         plt.pause(0.05)
41         continue
42
43     try:
44         value = float(raw)
45     except ValueError:
46         plt.pause(0.05)
47         continue
48
49     t = time.time() - start
50
51     times.append(t)
52     values.append(value)
53
54     maximum = max(values)
55     minimum = min(values)
56     average = sum(values) / len(values)
57
58     print(f"{t:.1f}s | {value:.3f} V | max {maximum:.3f} | min {minimum:.3f} | avg
59 {average:.3f}")

```

```
60 line.set_data(times, values)
61
62 ax.set_xlim(max(0, t - 60), t + 1)
63
64 info_text.set_text(
65     f"Aktuálně: {value:.3f} V\n"
66     f"Maximum: {maximum:.3f} V\n"
67     f"Minimum: {minimum:.3f} V\n"
68     f"Průměr: {average:.3f} V"
69 )
70
71 fig.canvas.draw_idle()
72 plt.pause(0.1)
```